MIL-S-20426
27 November 1951

MILITARY SPECIFICATION

SHELL, BASE EJECTION AND ILLUMINATING TYPE
(OTHER THAN MORTAR SHELL), METAL PARTS FOR

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-30-6
15 July 1948

This specification consists of this cover sheet, Specification 50-30-6, dated 15 July 1948, and Amendment 1, dated 1 February 1949, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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U. S. ARMY
SPECIFICATION

No. 50-30-6

Amendment-1
1 February 1949

SHELL, BASE EJECTION AND ILLUMINATING TYPE (OTHER THAN MORTAR SHELL), METAL PARTS FOR

The following changes shall be made in U. S. Army Specification No. 50-30-6, dated 15 July 1948:

Page 1, paragraph C-4. Add the following at the end of the paragraph:

"The interior cavity of each shell shall be cleaned by the shot or sand blast method, or by any other method approved by the Chief of Ordnance."

Page 1, paragraph C-6. Delete the entire last sentence.

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U. S. ARMY
SPECIFICATION

No. 50-30-6
15 July 1948

SHELL, BASE EJECTION AND ILLUMINATING TYPE (OTHER THAN MORTAR SHELL), METAL PARTS FOR

A. APPLICABLE SPECIFICATIONS.

A-1. The Ordnance Department drawings pertaining to the shell under contract and all drawings and specifications listed thereon shall form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of the invitation for bids.

B. TYPES AND CLASSES.

B-1. This specification covers the manufacture of the following types and classes of shell (other than mortar shell):

B-1a. Type A - Illuminating Shell.

B-1a(1). Class 1 - Shell manufactured with body of seamless steel tubing and adapter machined from steel casting.

B-1a(2). Class 2 - Shell manufactured by machining entire body of shell, including nose, from a one-piece forging.

B-1b. Type B - Base Ejection Shell.

B-1b(1). Class 1 - Shell manufactured with body of seamless steel tubing and adapter machined from bar stock or forging.

B-1b(2). Class 2 - Shell manufactured by machining entire body of shell, including nose, from a one-piece forging.

C. MATERIAL AND WORKMANSHIP.

C-1. All material and parts shall be in accordance with applicable specifications and drawings.

C-2. The Class 2 shell of both Types A and B shall be machined from forgings manufactured in accordance with U. S. Army Specification 57-104-2, Forgings For Common Steel Shell And Shrapnel.

C-3. The machining and treatment of the shell in each lot shall be uniform.

C-4. The interior of all shell shall be free from burrs, scale, fins, deep scars or pits, and sharp edges.

C-5. All shell shall be sound and free from cracks and porosity.

C-6. Cleaning Prior To Painting.---Projectile surfaces to be painted shall be subjected to Grade I treatment, Joint Army-Navy Specification JAN-C-490. These cleaned surfaces shall not be recontaminated prior to painting. After painting, these surfaces shall meet the various performance requirements of Joint Army-Navy Specification JAN-C-490. The interior cavity of each shell shall be cleaned by the shot or sand blast method, or by any other method approved by the Chief of Ordnance.

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C-7. Banding.—

C-7a. Band Seat.—Prior to banding, the band seats shall be clean and free from oil, grease, dirt, rust and other foreign material. Shot or sand blasting is not permitted for cleaning band seats after knurling.

C-7b. Rotating Band.—

C-7b(1). The rotating band blank shall be in the form of a solid ring, annealed if necessary to meet the requirements of U. S. Army Specification 50-27-1. The band blank shall be carefully pressed into the band seat so as to completely fill it and fit tight. (See Par. F-7).

C-7b(2). The band shall be applied after final heat treatment of the shell body. It shall be applied in such a manner as not to distort the walls of the shell. Distortion shall not be considered to have taken place if the band seat, and the shell body immediately adjacent to the band are within drawing tolerances specified.

C-7b(3). The bands shall be applied cold.

C-8. Protection Of Rotating Band.—The machined rotating bands shall not be nicked, burred, or otherwise damaged in subsequent handling and shipping.

C-9. Coating.—The exterior coating and the cavity coating specified on the drawing for the shell shall be applied in such a manner as to produce a tightly adherent coating of uniform thickness over the entire specified areas. The shell shall be stenciled or marked as prescribed on the drawing. All paint and marking shall be dry to the touch before shell are packed for shipment.

C-10. Body-Adapter Joint.—The flush or protruding edge of the adapter skirt shall be free from burrs or sharp edges. The forward edge of the body-tube shall not protrude beyond the adjacent edge of the adapter.

GENERAL REQUIREMENTS.

D-1. Forgings.—

D-1a. Forgings shall be grouped for heat treatment by lots which may be composed of forging lots, or (in later processing phases) miscellaneous forging lots.

D-1b. A forging lot shall contain no more than one heat of steel. Small lots (such as groups of abnormally small heats of steel or forgings which have become separated from previously delivered lots, hospitalized forgings, etc.) may be grouped into miscellaneous forging lots at the discretion of the machiner, and heat treated in such a manner as to obtain the specified minimum physical properties. Such lots shall be identified by a miscellaneous lot code number. The size of the miscellaneous forging lot shall not exceed 100 tons.

D-1c. During machining operations the steel heat code number of miscellaneous lot code number, shall be marked on each forging following any machining operation which removes it, so that the lot may be identified until it has passed final physical tests.

D-2. Shell requiring a nosing-in operation may be nosed-in either hot or cold. The nosing-in operation shall be made before the final heat treatment. No cold

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working involving plastic flow of metal is authorized after final heat treatment. The contractor shall establish a method satisfactory to the Chief of Ordnance, for maintaining the interior ogive contour of the shell.

D-3. Unless otherwise specified all shell, after the nosing-in operation and shell not required to be nosed-in, shall before finish machining be given a heat treatment wherein the shell is heated above the critical temperature range but not to exceed 1,600° F. and quenched in oil and drawn without delay at a temperature suitable to produce required physical properties but not less than 750° F. After removal from the draw furnace, shell shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700° F., after which cooling may be accelerated by a water spray or similar means. Except as modified by Paragraph F-4a, the heat treatment shall be uniform throughout for each lot and shall be so conducted as to produce in any shell subsequently selected by the inspector the physical properties specified on the shell drawing.

D-4. Point-fuzed shell may be annealed at the nose for threading provided the heating to redness does not extend more than one-half the distance from the nose to the bourrelet. A lead bath shall not be used for heating the nose without the specific approval of the Chief of Ordnance.

D-5. Base Plugs.—

D-5a. Base plugs shall be machined from forgings or flat bar stock.

D-5b. Base plugs shall be machined so that the axis of the finished plug shall be perpendicular to the axis of the original ingot.

D-5c. Base plugs shall have the physical properties specified on the drawing. For inspection and test they shall be divided into lots of the size specified for the shell to which they belong. Each lot of base plugs shall be of uniform chemical composition, and the machining and treatment of each lot of base plugs shall be uniform.

D-5d. Base plugs of base ejection shell shall be screwed firmly into the shell, but not so tightly as to make their removal difficult at the loading plant. All sharp edges on the base plug and at the junction of the base plug with the shell shall be removed.

D-5e. Base plugs of illuminating shell shall not be assembled into the shell by the manufacturer. All sharp edges on the base plug shall be removed.

D-5f. Base plugs of illuminating shell shall be manufactured from either electric furnace steel or open hearth steel and shall not contain more than .05 phosphorus nor more than .06 sulfur.

D-6. Seamless Steel Tubing.—Process lengths of seamless steel tubing being manufactured into shell bodies, may be routed through the machining plant without regard to heat identity, but the heat code symbol shall be retained on each body through final machining.

D-7. Adapters.—Each lot of adapters shall be of uniform chemical composition and the machining and treatment of each lot shall be uniform. All sharp edges of the adapters shall be removed.

D-8. Assembly Of Cover Discs.—The method of soldering cover discs over flash holes of adapters shall be such as to provide a moisture and air-proof seal. No solder shall be evident in flash hole or on surface of disc over flash hole after attachment of cover disc.

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DETAIL REQUIREMENTS.

E-1. Shell, base plugs, and adapters shall have the physical properties specified on the drawing.

E-2. The material of the rotating band shall show in test specimens taken from bands removed from the shell, the following physical properties:

Tensile Strength, lb. per sq. in., min.	
Shell to and including 155 mm	32,000
Shell larger than 155 mm	30,000
Percent Elongation, Min., in 1 in.	
Shell over 105 mm	25
Shell, 105 mm and smaller	15
Band Test Degrees	
Shell of all sizes	180

E-3. Band Seating.—After seating of rotating band, the clearance between band and band seat must not exceed .006".

METHODS OF SAMPLING, INSPECTION, AND TEST.

F-1. Inspection.—Inspection shall be as specified in U. S. Army Specification 50-0-1, General Specification For Ammunition Except Small Arms Ammunition, and Federal Specification QQ-M-151, Metals; General Specification For Inspection Of.

F-2. Lots And Lot Numbers.—

F-2a. Heat Treatment Lots.—After heat treatment, forged shell shall remain grouped according to forging lots or miscellaneous forging lots as provided by paragraph D-1b. Alternative design body-tubes shall remain grouped according to the lots in which they were heat treated. Each heat treatment lot thus constituted shall be the basis for the selection of samples for tests as hereinafter prescribed, except proving ground tests.

F-2b. Shell Lots.—Upon completion of all physical tests, the heat treatment lots shall be regrouped for future identity into shell lots to which will be assigned serial shell lot numbers. The maximum size of a lot shall be:

For sizes up to and including 105 mm	20,000 shell
For sizes above 105 mm	10,000 shell

Each shell lot as above constituted shall be the basis for selection of samples for proving ground tests. The size of the pilot lot shall be equal to the greatest number of ballistic samples prescribed for that particular item.

F-2c. Serial numbers for shell lots shall begin with the number "1" and shall continue serially for each contractor in such a way that there will never be a duplication of numbers used by the same contractor on the same caliber of shell. This shell lot serial number shall appear stamped on the finished shell as prescribed in U. S. Army Specification 50-0-1.

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F-3. Hardness Test.—To insure uniformity of heat treatment, a hardness test with the Brinell machine using five (5) seconds minimum dwell, or by any other method approved by the Chief of Ordnance shall be made near the base of every shell. Each decrementally hardened shell shall also be tested near the bourrelet.

F-4. Tension Tests.—

F-4a. Two shell, or process lengths of tubing as indicated, one of the hardest and one of the softest, as determined by the hardness test, shall be selected by the inspector from each forging lot, (or from the first 20% of each heat of tubing), for tension tests. At the option of the contractor a production control sample may be utilized, the number of shell in the production control sample to be determined by the contractor. When a production control sample is utilized, the hardness range thus established shall be maintained throughout the balance of the heat treatment of that lot. Shell which fall without the established hardness range may be reheat-treated to fall within the originally established range; or else new physical properties may be determined from newly selected shell or body-tubes, which represent the true extremes of hardness. Double the prescribed number of forged shell shall be selected as above from each miscellaneous forging lot. When shell bodies are differentially hardened, tension test specimens conforming to Type 1, Federal Specification QQ-M-151, or (if the wall thickness of the shell is not sufficient for Type 1) to the largest size of Type 4 specimen obtainable, shall be taken longitudinally, two from each of the opposite sides of the wall of each of the sample shell. Four tension test specimens are thus selected from each shell. All test specimens may, if desired, be taken from a longitudinal half-section of the shell provided that each specimen is taken adjacent to the sectioned face of the half-section. In other words, the specimens may be taken from locations which are somewhat less than 180° apart. Two test specimens shall be taken in such position that the gage length will include the section of the wall at the location of the band seat, and in such a manner that the test coupon is from within the contours of the finished shell. The remaining two specimens shall be selected so that the gage length is forward of the transition zone. When shell bodies are uniformly hardened, only two specimens need be taken, but the gage length of both shall include a section of the wall at the location of the band seat.

F-4b. Two base plugs, or a section from two bars used in the production of base plugs, shall be selected by the inspector from each lot (see Par. D-5c) for tension tests. One tension test specimen conforming to Type 1, Federal Specification QQ-M-151, or (if the plug is too small) to the largest size of type 4 specimen obtainable, shall be cut from each plug in such a manner that the longitudinal axis of the specimen will be parallel to direction of rolling or forging.

F-4c. The yield strength shall be determined by either the offset method or by the extension-under-load method, as prescribed in Federal Specification QQ-M-151. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length). In addition the yield point shall be determined by the divider method described in Federal Specification QQ-M-151. The results shall be recorded but shall not be basis for acceptance of the shell.

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F-4d. Retests.--In case of failure to meet physical requirements, a retest of specimens from double the number of shell or base plugs of approximately the same hardness as the shell or base plug from which the failing specimen was taken, shall be made upon request of the contractor.

F-4e. Reheat-treatment And Resubmission.--Any heat treatment lot of shell or lot of base plugs that fail to meet the physical requirements may be reheat-treated and resubmitted for test.

F-6. Soundness.--

F-6a. Each shell shall be inspected for cracks and porosity after finish machining. All shell which show cracks or porosity shall be rejected. Doubtful shell shall be rejected except that at the option of the contractor they may be hydrostatically tested as provided in Paragraph F-6b.

F-6b. If the contractor elects to test doubtful shell hydrostatically, he shall provide hydraulic equipment capable of testing shell to an internal pressure of 25,000 pounds per square inch. Each doubtful shell shall be subjected to hydraulic pressure sufficient to stress the metal of the body to the prescribed yield strength of the material. The hydraulic pressure to be used shall be determined by the following formula:

$$P = \frac{2St}{D}$$

where
P = pressure in pounds per sq. in.
S = fiber stress in pounds per sq. in.
t = maximum allowable wall thickness at forward bourrelet.
D = bourrelet diameter in inches.

F-7. Rotating Band Tests.--

F-7a. Detailed Inspection.--

F-7a(1). Method Of Inspection For Shell Other Than 105 MM, M1 and M60.--One band per 2,000 shell from each banding machine (or where production is less than 2,000 per working shift, one band from each banding machine from each shift) shall be removed for examination of the seating. In case of 8" and 240 mm shell one band shall be removed for examination of the seating from one shell per banding machine per 24 hour day, selecting the sample from each shift in rotation. Prior to removal, the diameter of the finished band shall be measured on at least three diameters 60° apart and on at least two points along the length of the cylindrical portion, near the front and near the rear (a minimum of six diametral measurements). These points of measurement shall be marked on the bands as well as the orientation of the band on the shell. After removal, the thickness of the band shall be measured at each of the marked points (twelve measurements) and the diameter of the band seat shall be measured opposite those points. Add to the diameters of the band seat at each point, the corresponding sum of the two thicknesses of the band and subtract the result from the corresponding exterior diameter of the band previously determined. The

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result will indicate the total diametral clearance between the band and its seat. If the band is well seated, a negative clearance or interference may be indicated on some of the measurements or even on the average of all measurements. In case the indicated clearance thus determined is greater than .006 inch on any one diameter, either front or rear, the band shall be removed from ten additional shell from the group represented. If a clearance greater than .006 inch on any one diameter is indicated on any one of these additional shell, the entire represented shall be rejected subject to rebanding and retest. When waved band seat ribs or irregular band profile interferes with ease of measurement, the manufacturer is authorized to machine the outside of the band in order to obtain an adequate cylindrical surface. The diameter of the machined band must exceed the bourrelet diameter.

F-7a(2). Method Of Inspection For 105 MM Howitzer Shell.--A hydraulic press having one fixed and one moveable ram, capable of exerting a total ram pressure of 12,000 lb. min. ram pressure shall be used for this test. Two indenters (or anvils) of rectangular cross section, .378" by .378", each having a face radius of 2.11" (to match band diameter) shall be provided, one on the fixed and one on the moveable ram of the press, but both on a common center line. The press shall be fitted with dial indicators to show pressure on ram and motion of ram. (See Par. H-2).

The pressure used (approx. 10,000 lb. ram pressure) shall be great enough to seat the band under the indenters, but not so great that permanent deformation of the shell occurs.

The band seating shall be inspected by pressing the band between the indenters in five planes approximately 36° apart and measuring the diameter of the rotating band in each plane before and after application of the pressure. Where the press is equipped with a dial indicator to measure the diameter across the impressions, care must be taken not to take the reading until after the full pressure has been applied, and the motion of the ram has ceased as shown by the dial indicator.

The impressions left by the indenters shall not be cause for rejection of the shell. The difference in these two diametrical readings will be the sum of the clearances under the band at two points in one plane. The sum of the clearances under the band in any one plane shall not be greater than .006". The average indicated clearance which is the sum of the readings taken in all five planes divided by ten shall not be more than .002".

This test shall be performed by the contractor at his discretion to assure adequate process control of band seating. In addition the test shall be performed by the inspector on a representative sample from each lot or sub-lot; if the sample reveals unsatisfactory banding, the entire lot or sub-lot will be rejected subject to 100% inspection or rebanding and resubmission for all acceptance tests.

F-7a(3). Character Of Seating.--Each band removed in accordance with Paragraphs F-7a(1), F-7b and F-7c shall be examined for character of seating and to determine whether corners of the band seat have been properly filled. (See Par. F-8a).

F-7b. From each 2,000 shell a sufficient number of unmachined bands, or bands from which a light surface cut (.02 maximum) has been taken, shall be removed to

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provide one specimen to be subjected to each of the physical tests prescribed in Paragraph E-2. Whenever practicable, bands removed for the seating test may be used, and specimens for both tension test and bending test may be taken from the same blank. The bands shall be carefully removed from the shell and carefully flattened without hammering. Each test specimen shall be cut along the direction of the flattened length and prepared in such a manner as to hold to the minimum any additional cold working. Tension test specimens shall be machined to the form and dimensions prescribed in U. S. Army Specification 50-27-1. For tensile test of bands less than .5 inches in width use a specimen machined to the maximum rectangular cross-section which can be obtained from an unmachined band.

F-7c. In the bend test, each band or section thereof, of full width and thickness except as reduced by authorized surface cut, shall be bent cold by pressure, without hammering, in the direction of curvature of the band, until it tightly incloses 180° of a rod equal in diameter to the thickness of the bend test specimen (within ± .01").

F-7d. If any specimen fails to meet the tensile or elongation requirements of the physical test, or if any band cracks in the bend test, a retest may be made at the request of the contractor. Double the number of samples shall be used for the retest and if failure occurs in the retest, the group of shell represented by the sample shall be rebanded and resubmitted for test.

F-8. Proving Ground Tests.—

F-8a. Proving ground tests for recovery, range, and velocity are not acceptance tests, but the results will be scrutinized carefully by the Ordnance Department to determine quality of the ammunition produced. Failure in any phase of the proving ground tests will result in reinspection of the lot by the Ordnance Department. If the reinspection reveals that latent defects were responsible for the failure, the lot will be rejected. Improper seating of the rotating band will be considered to be a latent defect. (See Par. F-7).

F-8b. For recovery firing, velocity and range tests (See Pars. F-8d and F-8e) the inspector shall select samples as follows:

Ten (10) representative shell will be selected from each completed shell lot, stamped "R" and shipped to the proving ground. These are to be fired for recovery. At the completion of ten (10) consecutive successful firings, the sample will be reduced to one shell selected from each shell lot. These will be stamped "R" and when ten (10) shell have been accumulated, they will be shipped to the proving ground for recovery firing. In addition, and except as hereinunder provided, one (1) representative shell will be selected from each 10 percent of each shell lot for range and velocity firing. These will be stamped "V" and immediately upon completion of the shell lot, will be shipped to the proving ground. In all cases samples forwarded to the proving ground shall be identified as to the lot they represent.

F-8c. At the proving ground, the sample shell assemblies stamped "R" shall be loaded with standard loaded filler components required by the loading drawing, except that four (4) of the shell shall be assembled with inert charges and inert fuzes, and the remaining six (6) shell assembled with live fuzes and standard

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black powder bursting charges. All ten (10) shell shall be fired for recovery from a weapon for which the shell is standard, using a propellant charge that will give 12 percent excess pressure over the maximum rated pressure for the weapon. The six (6) rounds which have live fuzes and live burster charges shall have the fuzes set for air bursts.

F-8c. At least 3 of the 4 shell which were assembled with inert fuzes and inert burster charges, and at least 2 of the 6 shell which were assembled with live fuzes and black powder charges, shall be recovered and examined to determine whether upsetting occurred in the bore (as evidenced by rifling marks extending more than 180° around the body) or by an increase or decrease in the diameter of the shell body at the bourrelet, and forward and behind the rotating band, whether the rotating bands took the rifling properly, whether dishing or upsetting of the base plugs occurred, whether the shell assemblies were cracked or broken. Also to determine, of the shell assembled with live fuzes and live burster charges only, whether the black powder chambers remained unruptured and assembled to the shell bodies and whether the base plugs and metal smoke units were ejected satisfactorily; and of the shell assembled with inert fuzes and inert burster charges only, whether the base plugs remained attached to the shell bodies.

F-9. Assembly Of Adapters.—Each Class 1 shell of both Types A and B shall be subjected to an internal pressure of 15 p.s.i. minimum for 15 seconds minimum. Means shall be provided to prevent the pressure from being exerted on the cover disc of Type B assemblies. There shall be no evidence of leakage through the body-adapter joint.

F-10. Assembly Of Cover Disc.—

F-10a. After soldering of cover disc to shell body or adapter, and before painting, shell body assemblies or adapter assemblies shall be visually examined to determine that there is no solder on the fuze side of the disc within approximately 3/16" of center of the disc, nor on the flash hole side of the disc nor in the flash hole.

F-10b. After the test of adapter tightness (See Par. F-9), each shell assembly having a cover disc (both Classes of Type B shell assemblies) shall be subjected to an air pressure of five (5) p.s.i. minimum for 15 seconds minimum applied through the fuze hole. There shall be no evidence of leakage through the flash hole.

F-10c. Shell rejected in accordance with Paragraphs F-9, F-10a and F-10b above may be reworked by the contractor and shall be accepted if they fulfill all requirements of the drawing and specification, including Paragraphs F-9, F-10a and F-10b above.

G. PACKING AND MARKING.

G-1. Packing.—

G-1a. Shell Smaller Than 75 MM, shall be packed in commercial type containers so constructed as to insure acceptance by common or other carriers for safe transportation at the lowest rate to the point of delivery.

G-1b. Shell 75 mm to 105 mm, inclusive, shall be packed on pallets as specified on Drawings 76-18-5, 76-18-5A, 76-18-7 and 76-0-48.

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G-1c. Shell larger than 105 mm shall be provided with grommets of approved type and eye-bolt lifting plugs, and shall be packed on pallets as specified on the following drawings:

155 mm shell --------	76-18-14	8 inch shell --------	76-18-17
6 in shell --------	76-18-12	240 mm shell --------	76-18-19

G-1d. Palletized or packaged shell shall be loaded in closed box cars or other closed carriers as directed by the contracting officer and so as to insure acceptance by common or other carrier.

G-1e. Data Cards.--Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted projectiles to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

AMMUNITION DATA CARD

Kind_____ Lot No._____
Quantity in Lot_____
Drg. No.*_____ Spec. No.*_____
Manufactured by_____ Date_____
Contract No._____ Order No._____
Packed_____
Remarks_____

Certified to by_____, Inspector
(Print here name of inspector, district, arsenal, or loading plant, and sub-district, where applicable)

*Include revision and amendments.

G-2. Marking.--The marking for shipment shall be in accordance with U. S. Army Specification 100-2, Standard Specification For Marking Shipments By Contractors.

H. NOTES.

H-1. The necessity for firmly seating the rotating band so that it is in contact with the shell at all points cannot be over emphasized. Air gaps of even a few thousandths of an inch result in differences in starting resistance when the band enters the rifling. This causes variations in the burning characteristics of the powder which in turn affects the muzzle velocity and finally the range. Bands should be fully and uniformly annealed prior to seating and pressure should be adequate and uniformly applied in order to obtain satisfactory results.

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H-2. The Detroit Testing Machine Model HB-1 is suitable for the test prescribed in Paragraph F-7a(2).

H-3. This specification replaces Ordnance Department Tentative Specifications AXS-877 and AXS-1545.

H-4. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

NOTE: Copies of this specification may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

(No. 50-30-6)

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